CYCLISTIC BIKE

Analysis report

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Background

Cyclistic is a bike-share company based in Chicago. The company offers flexible pricing plans, including single-ride passes, fully-day passes, and annual memberships. Customers who purchase single-ride or full day passes are classified as casual riders, while customers who purchase annual memberships are classified as members.

Cyclistic's finance team has determined that annual members are significantly more profitable than casual riders. As a result, the marketing team aims to increase the numbers of annual memberships by converting casual riders into members.

Business Task

Analyze Cyclistic's historical trip data to identify how casual riders and annual members use the bike-share service differently.

Key Stakeholders

- Lily Moreno
- Cyclistic Marketing Analysis Team
- Cyclistic Executive Team

Business Objective

The objective of this analysis is to uncover behavioral differences between casual riders and annual members and provide data-drien recommendations to help Cyclistic convert more casual riders into annual members.

Process

During the processing stage, the 12 monthly csv files were loaded and merged into a single DataFrame using Python and Pandas library.

The dataset structure was inspected using **columns** and **info()** to understand the available fields and data types.

The **started_at** and **ended**_at columns were converted to datetime format. Additional features were created to support the analysis, including:

- **ride_length**: during of each ride
- **day_of_week**: name of the day on which the ride started
- **month**: month of the ride
- **hour**: hour when the ride started

The dataset was checked for duplicate records, and no duplicates were found.

Missing values were identified in station-related columns and ending coordinates. Rows containing missing values in **start_station_name**, **start_station_id**, **end_station_name**, **end_station_id**, **end_lat** and **end_len**. **end_lat** and **end_len** were removed to improve data completeness, and the cleaned dataset was saved as **df_processed.csv**

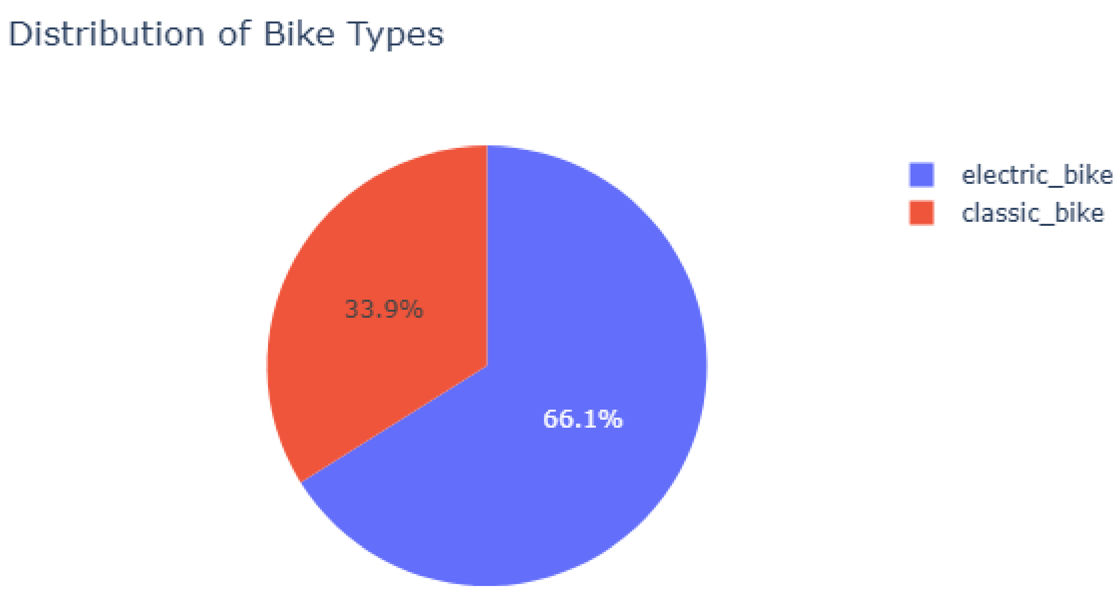
In addition, a secondary dataset named **df_without_nulls** was created by removing all remaining missing values. This dataset was used for analyses that required complete station information such as identifying the most frequently used start and end stations.

Analysis

Rideable type usage

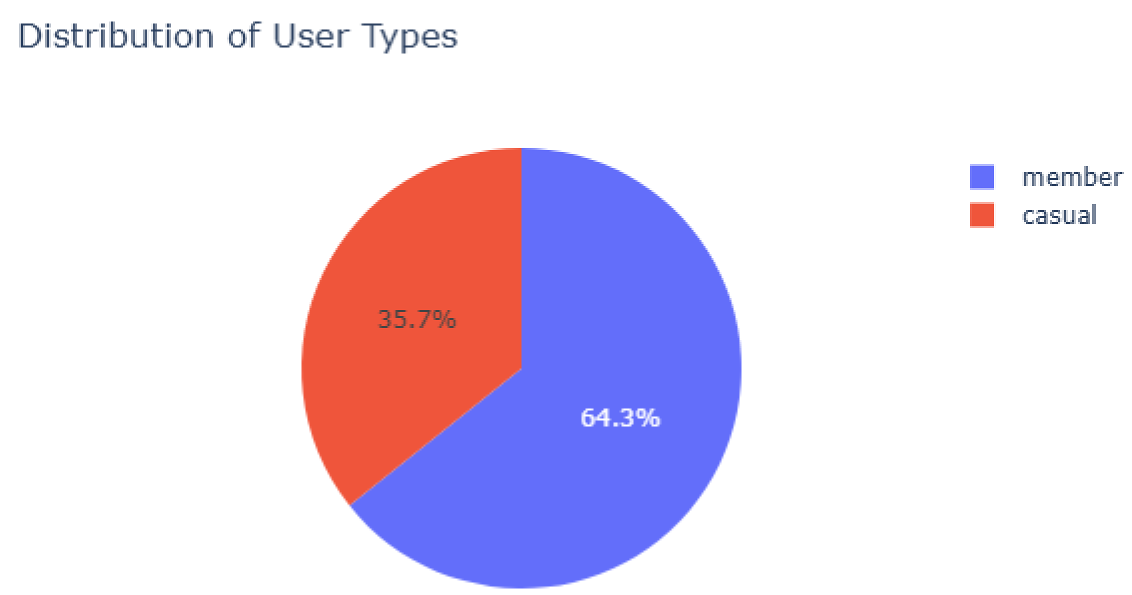
Analysis of rideable bike preferences shows that electric bikes were used significantly more than classic bikes. Electric bikes recorded 66.1%, compared to 33.9% for classic bikes.

This suggests that electric bikes are more attractive to rides and play an important role in overall bike usage.



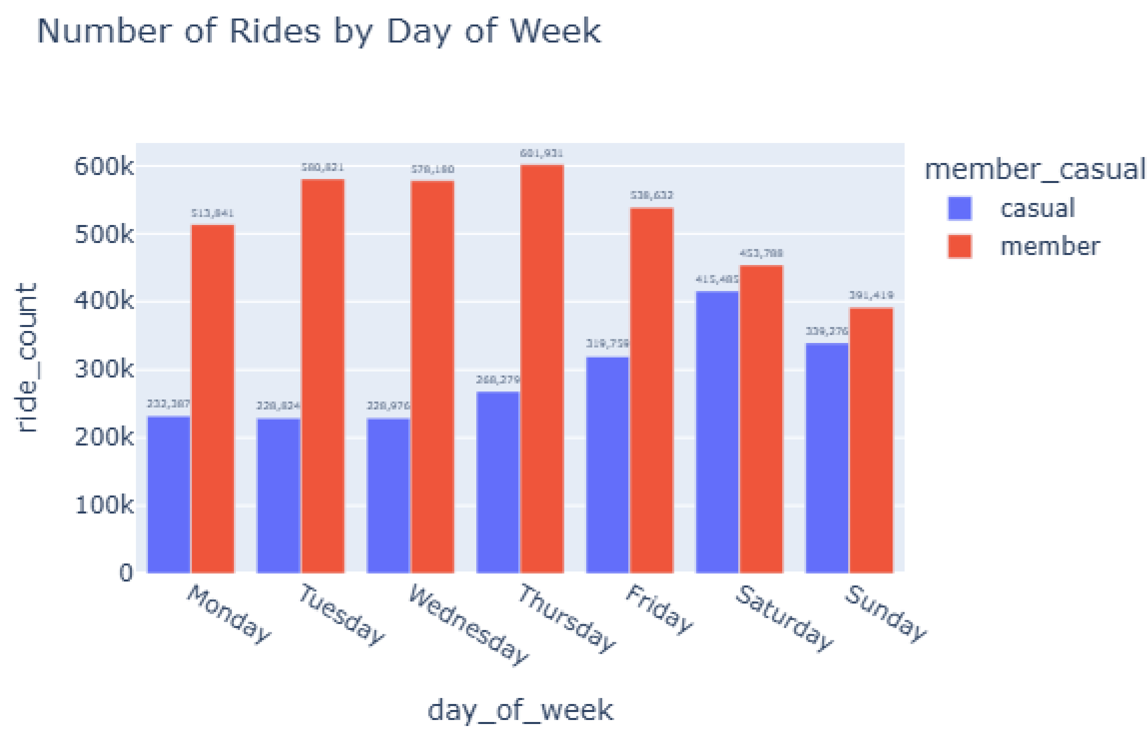
User type percentage

Members make up 64.3% of the total users in the dataset, compared to 35.7% for casual users. This indicates that most bike usage comes from registered members rather than casual riders.



Weekly usage

Casual users are most active during weekends, especially on Saturday, while members are more active during weekdays, with peak usage on Thursday. The difference suggests that casual riders mainly use bikes for leisure, whereas members use them more for daily commuting.

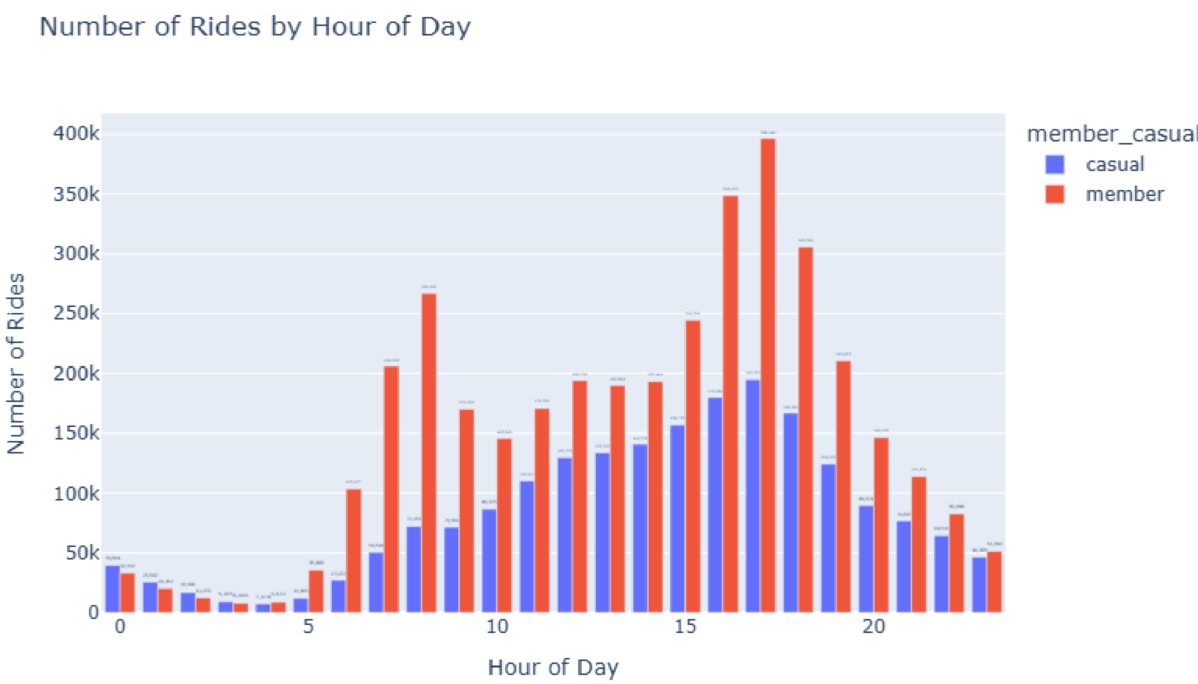


Hourly usage

The analysis shows that both casual users and members are most active during the late afternoon and early evening hours, especially around **5 PM (17:00).**

For casual users, ride activity gradually increases during the afternoon, peaking between **4 PM and 6 PM**, which may indicate recreational or social riding behavior. Activity remains relatively lower during early morning hours.

Members also show peak usage around **5 PM**, but they demonstrate noticeably higher activity during commuting hours, particularly around **7 AM to 9 AM** and late afternoon hours. This pattern suggests that members commonly use the bike-sharing service for work or daily transportation.

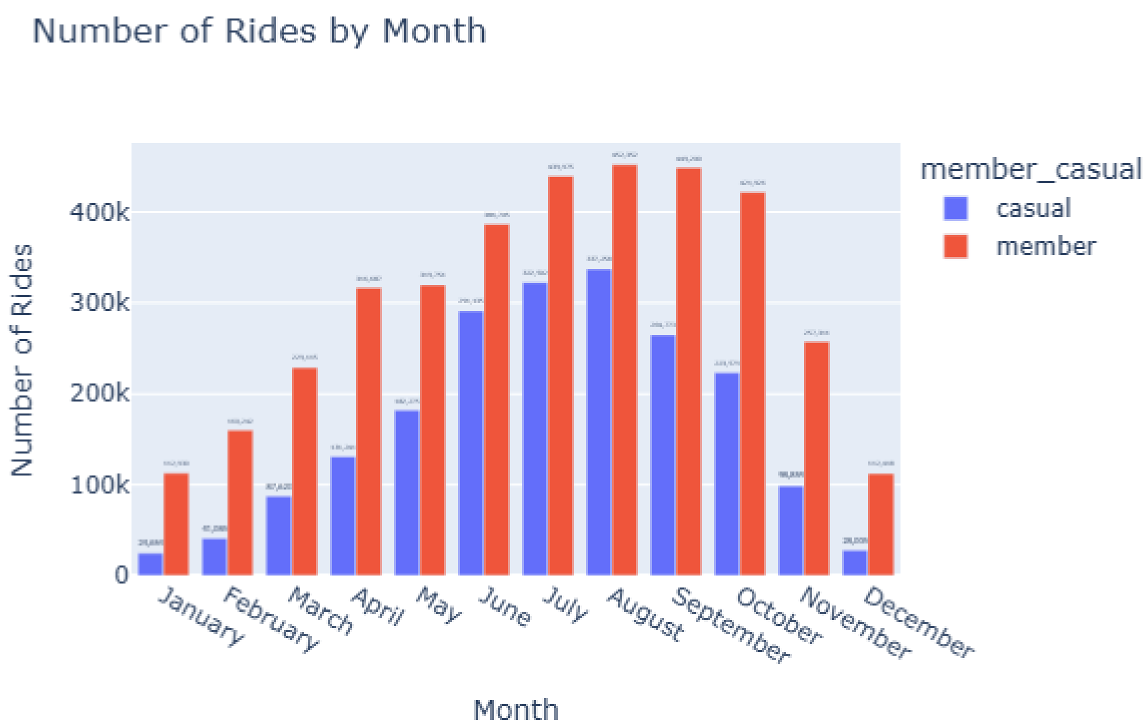


Monthly riding trends

The analysis shows clear seasonal patterns in bike usage for both casual users and members.

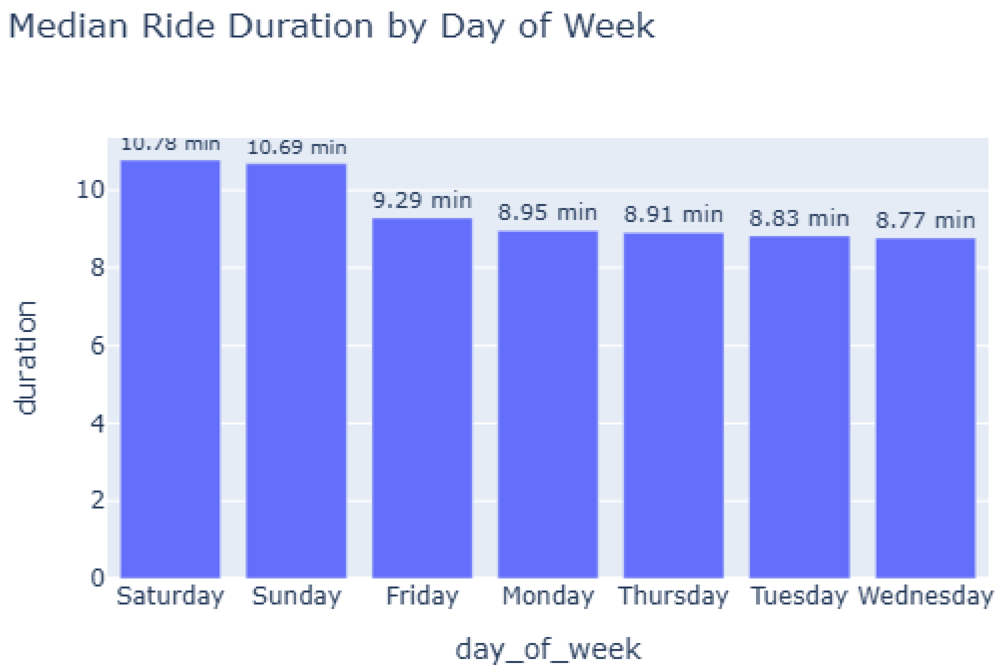
Casual users recorded the highest number of rides during **August** and **July**, while usage significantly decreased during winter months such as December and January. This suggests that casual riders are more influenced by weather conditions and tend to use the service mainly during warmer seasons for leisure activities.

Members also showed peak activity during the summer and early fall months, especially in **August, September, and July.** However, member usage remained relatively more stable throughout the year compared to casual users, indicating that members rely on the bike-sharing service more consistently for daily transportation**.**



Average ride duration

The analysis shows that casual users have a slightly longer average ride duration than members. Casual riders spend an average of 11.7 minutes per ride, while members average 10.03 minutes per rides. The suggests that casual users may use the bike-sharing service more for leisure or recreational purposes, whereas members tend to take shorter and more routine trips, likely related to commuting or daily transportation.

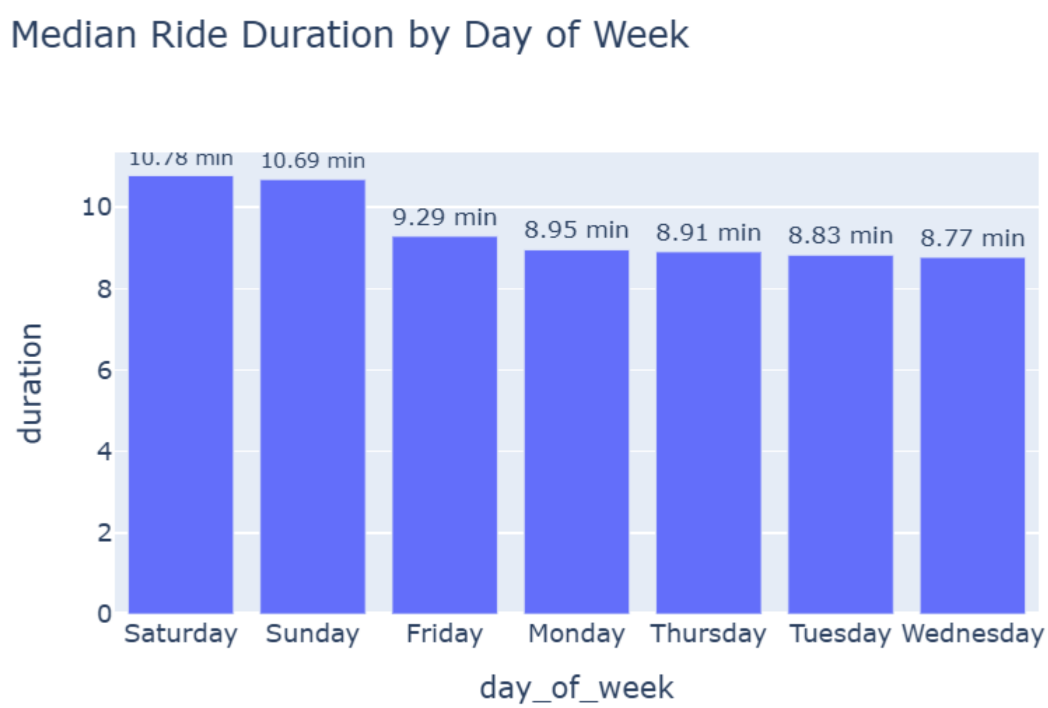


Average duration by day

The analysis shows that ride duration varies slightly across the week. The longest average ride durations occur on **Saturday (10.78 minutes)** and **Sunday (10.69 minutes).**

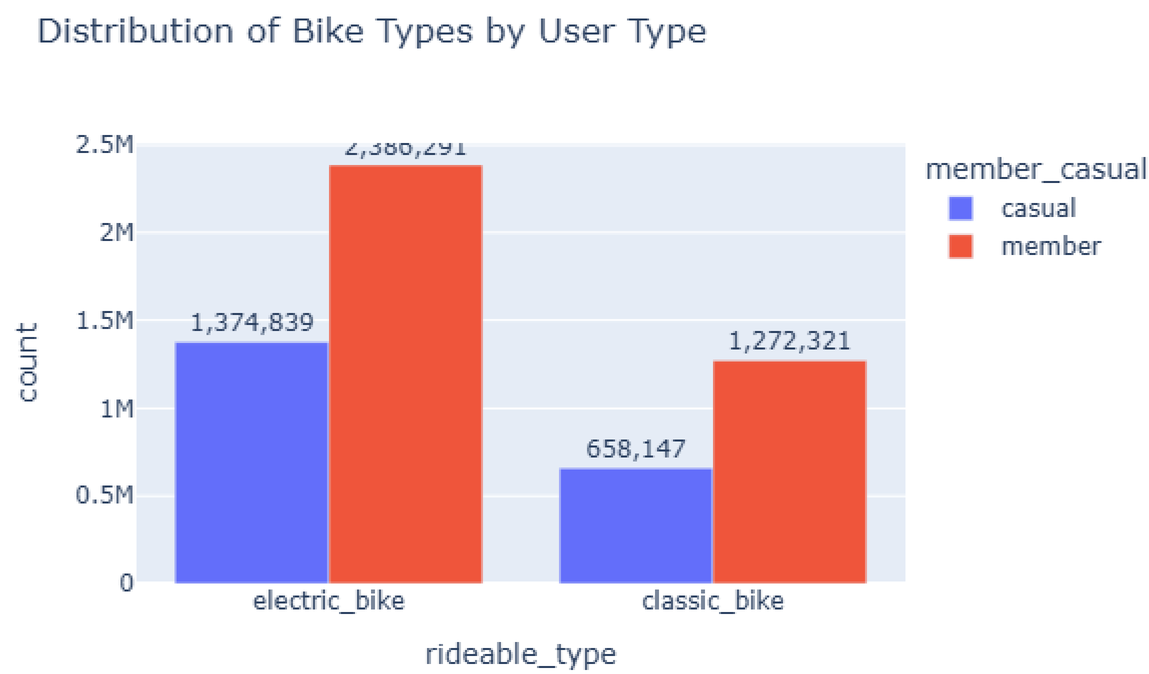
In contrast, weekday show shorter average durations, with the lowest values recorded on **Wednesday (8.77 minutes)** and **Tuesday (8.83 minutes).**

This suggests that users tend to take longer rides during weekends, likely for leisure or recreational activities, while weekday rides are shorter and more likely associated with commuting or routine travel.



Bike type by user

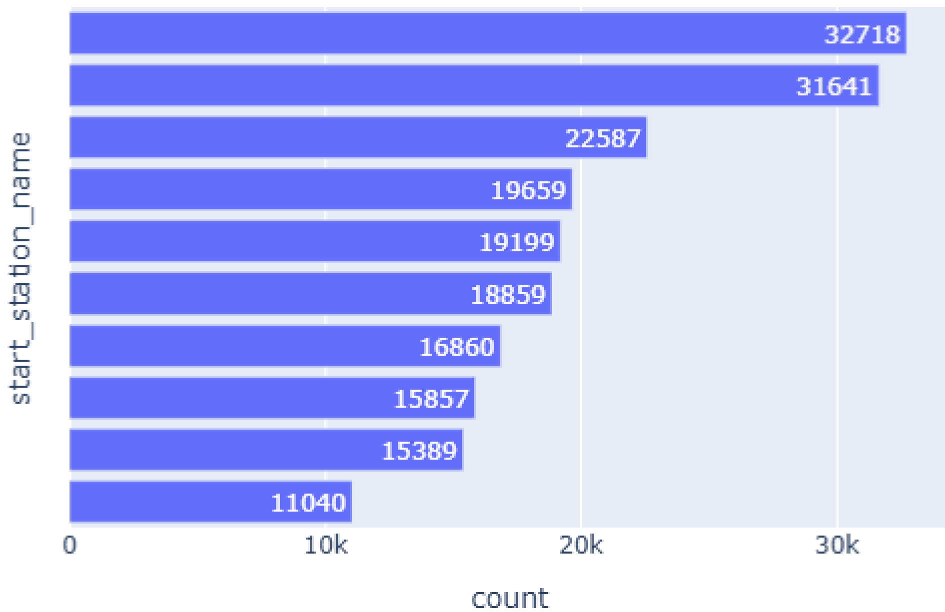
The analysis shows clear differences in bike preferences between casual users and members. For casual users, electric bikes are more popular, with 1,374,839 rides, compared to 658,147 rides for classic bikes. This suggests that casual riders prefer electric bikes, likely due to their ease of use and lower physical effort. Similarly, members also prefer electric bikes, recording 2,386,291 rides, while classic bikes account for 1,272,321 rides. However, members show a stronger overall usage of both bike types compared to casual users. Overall, electric bikes are the most preferred option for both user groups, indicating that they play a key role in the bike-sharing system.



Most popular start stations for casual users

The analysis identifies several highly popular start stations among casual users, with many located near tourist attractions, parks, and recreational areas.

The most frequently used start station in Navy Pier with 32,718 rides, followed by DuSable Lake Shore Dr & Monroe St with 31,641 rides, and Michigan Ave & Oak St with 22,587 rides. Other popular locations include Streeter Dr & Grand Ave, Millennium Park, and Shedd Aquarium. These findings suggest that casual users primarily use the bike-sharing service in leisure and tourist-oriented locations rather than for daily commuting. This behavior supports the idea that casual riders are most associated with recreational usage patterns.

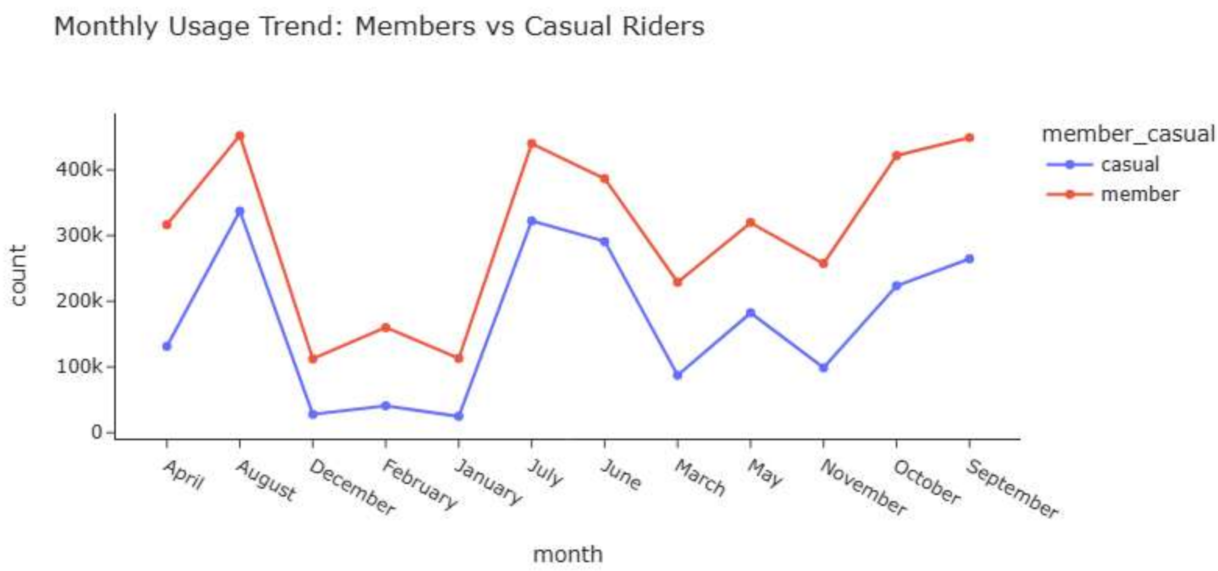


Seasonal usage trends

The analysis shows that bike usage changes significantly across different months for both casual users and members.

Ride activity increases steadily during warmer months and reaches its peak in July, August, and September. August records one of the highest usage levels for both casual users (337,258 rides) and members (452,352 rides).

In contrast, usage drops sharply during winter months such as December and January, especially among casual users. This indicates that weather and seasonal conditions strongly influence bike-sharing activity. Additionally, members maintain higher ride counts throughout the year compared to casual users, suggesting that members depend on the service more consistently, while casual users are more seasonal and leisure and leisure-oriented.



Recommendations and Key Insights

Based on the analysis, the following recommendations can help increase customer engagement and encourage casual users to become annual members:

- **Target casual users with weekend promotions:**
  - Since casual are most active on weekends, the company can introduce weekend membership discounts, free trial memberships, or special ride packages to encourage conversion to annual memberships.
- **Focus marketing on tourist and recreational areas:**
  - Popular casual-user stations such as Navy Pier Millennium Park, and Shedd Aquarium indicate strong leisure usage. Marketing campaigns, digital advertisements, and membership offers can be placed in these high-traffic tourist locations.
- **Increase electric bike availability:**
  - Electric bikes are the most preferred rideable type for both casual users and members. Expanding electric bike availability and ensuring proper maintenance could improve customer satisfaction and increase ridership.
- **Seasonal marketing campaigns:**
  - Usage peak during summer months, especially July and August. The company should launch seasonal campaigns, promotions, and outdoor events during high-demand periods to maximize engagement and revenue.
- **Encourage membership through customer benefits**

    - Members show strong weekday commuting behavior. The company can promote memberships by highlighting benefits such as cost savings, convenience, and unlimited rides for daily commuters.
- **Optimize bike distribution by time and location**
    - Since ride activity peaks during afternoon and evening hours, especially around 5 PM, the company should ensure bike availability during peak periods and at high-demand stations.
- **Personalized marketing strategies:**
    - Casual users and members show different riding behaviors. Creating separate marketing strategies for each group could improve customer targeting and increase conversion rates.